

MIAE 221 · Rapid Review Sheet · Part 3

Crystal Structures, Unit Cells, Theoretical Density & Miller Indices

1. Metallic Unit Cell Crystal Geometries

Structure	Coordination No. (CN)	Atoms/Cell (n)	Lattice Parameter a(R)	Close-Packed Direction	APF	Stacking Sequence	Examples
Simple Cubic (SC)	6	1	a = 2R	Cube edges <100>	$\frac{\pi}{6} \approx 0.52$	AAAAA...	Polonium (α-Po)
Body-Centered Cubic (BCC)	8	2	$a = \frac{4R}{\sqrt{3}}$	Body diagonal <111>	$\frac{\pi\sqrt{3}}{8} \approx 0.68$	ABABAB...	α-Fe, Cr, W, Mo, Ta
Face-Centered Cubic (FCC)	12	4	$a = 2\sqrt{2}R$	Face diagonal <110>	$\frac{\pi\sqrt{2}}{6} \approx 0.74$	ABCABC... (close-packed {111})	Cu, Al, Au, Ag, Ni, γ-Fe
Hexagonal Close-Packed (HCP)	12	6 (2 in prim.)	$a = 2R, c/a = 1.633$	Basal plane {0001}	0.74	ABABAB... (close-packed basal)	Ti, Mg, Zn, Be, Cd, Zr

2. Theoretical Density Formula & Unit Conversions

ρ = (n · A) / (Vc · NA)

- Parameters: n = atoms/unit cell (SC = 1, BCC = 2, FCC = 4, HCP = 6), A = atomic mass (g/mol), Vc = unit cell volume (cm³), NA = 6.022 × 10²³ atoms/mol.
- Cubic Volume: Vc = a³. For BCC: Vc = (4R/√3)³ = 64R³/3√3. For FCC: Vc = (2R√2)³ = 16√2R³.
- Critical Conversions: 1 nm = 10⁻⁷ cm ⇒ 1 nm³ = 10⁻²¹ cm³; 1 Å = 10⁻⁸ cm ⇒ 1 Å³ = 10⁻²⁴ cm³. Target density units: g/cm³.

3. Seven Crystal Systems & 14 Bravais Lattices

System	Axial Lengths	Interaxial Angles	Bravais Lattices
Cubic	a = b = c	α = β = γ = 90°	Simple, Body-Centered, Face-Centered (3)
Tetragonal	a = b ≠ c	α = β = γ = 90°	Simple, Body-Centered (2)
Orthorhombic	a ≠ b ≠ c	α = β = γ = 90°	Simple, Body-Centered, Face-Centered, Base-Centered (4)
Rhombohedral (Trigonal)	a = b = c	α = β = γ ≠ 90°	Simple (1)
Hexagonal	a = b ≠ c	α = β = 90°, γ = 120°	Simple (1)
Monoclinic	a ≠ b ≠ c	α = γ = 90°, β ≠ 90°	Simple, Base-Centered (2)
Triclinic	a ≠ b ≠ c	α ≠ β ≠ γ ≠ 90°	Simple (1)

4. Crystallographic Directions [uvw] & Planes (hkl)

- Directions [uvw]: Vector from tail (x₁, y₁, z₁) to head (x₂, y₂, z₂) ⇒ Δx = x₂ - x₁, Δy = y₂ - y₁, Δz = z₂ - z₁. Divide by a, b, c. Reduce to smallest integers ⇒ [uvw]. Negative index denoted by overbar (e.g. $\bar{1}10$). Family of symmetrically equivalent directions: <uvw>.
- Planes (hkl) (Miller Indices):
 - Origin Check: If plane passes through origin, shift origin to an adjacent corner along a unit cell edge.
 - Intercepts: Determine intersection points with x, y, z axes in units of a, b, c (if parallel, intercept = ∞).
 - Reciprocals: Take 1/intercept (1/∞ → 0).
 - Enclose: Clear fractions to obtain smallest integers and enclose in parentheses (hkl). Family of equivalent planes: {hkl}.
- Isotropy vs Anisotropy: Single crystals are **anisotropic** (properties like modulus E vary with crystallographic direction). Polycrystalline materials with randomly oriented grains are **isotropic** (uniform macro properties).